

Solving Problems Involving Linear Functions

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A **linear function** is a function that can be written in the form $y = mx + b$, where m is the **slope** and b is the **y-intercept**.

Slope

The **slope** represents the rate of change between two quantities.

Y-Intercept

The **y-intercept** represents the initial value when the independent variable is zero.

Solving Word Problems

To solve a word problem:

Identify the changing quantity and the starting value.

Write a linear function that represents the situation.

Substitute the given value into the function.

Interpret the answer in the context of the problem.

Check Your Answer

Always check whether the answer is reasonable based on the given situation.

Examples

Example 1

A taxi charges a base fare of ₦45 plus ₦12 per kilometer traveled. Find the fare for a trip of 8 kilometers.

Example 1

Write the linear function:

$$y = 45 + 12x$$

For $x = 8$:

$$y = 45 + 12(8)$$

$$y = 45 + 96 = \text{P}141$$

Therefore, the fare is

$$\boxed{\text{P}141}.$$

Example 2

A printing shop charges a setup fee of ₱150 and ₱4 for each page printed. Find the cost of printing 60 pages.

Example 2

Write the linear function:

$$y = 150 + 4x$$

For $x = 60$:

$$y = 150 + 4(60)$$

$$y = 150 + 240 = \text{P}390$$

Therefore, the cost is

$$\boxed{\text{P}390}.$$

Example 3

A cyclist has already traveled 10 km and continues at a rate of 18 km per hour. Find the distance after 5 hours.

Example 3

Write the linear function:

$$y = 10 + 18x$$

For $x = 5$:

$$y = 10 + 18(5)$$

$$y = 10 + 90 = 100$$

Therefore, the distance is

$$\boxed{100 \text{ km}}.$$

Example 4

A phone plan costs ₱300 per month plus ₱2 for every text message sent. Find the bill if 250 text messages are sent.

Example 4

Write the linear function:

$$y = 300 + 2x$$

For $x = 250$:

$$y = 300 + 2(250)$$

$$y = 300 + 500 = \text{P}800$$

Therefore, the bill is

$$\boxed{\text{P}800}.$$

Practice Exercises

Practice 1

A taxi charges a base fare of ₱45 plus ₱12 per kilometer traveled. Find the fare for a trip of 8 kilometers.

Practice 2

A printing shop charges a setup fee of ₱150 and ₱4 for each page printed. Find the cost of printing 60 pages.

Practice 3

A cyclist has already traveled 10 km and continues at a rate of 18 km per hour. Find the distance after 5 hours.

Practice 4

A phone plan costs ₱300 per month plus ₱2 for every text message sent. Find the bill if 250 text messages are sent.

Practice 5

A student starts with a score of 50 points and earns 6 additional points for every completed task. Find the score after completing 20 tasks.

Activities

Activity 1

Carlo has already saved ₱650 and adds ₱80 each week. How much will he have after 8 weeks?

Activity 2

A water tank contains 150 liters of water and is being filled at a rate of 12 liters per minute. How much water will there be after 15 minutes?

Activity 3

A gym membership requires a registration fee of ₱900 and a monthly fee of ₱300. Find the cost after 10 months.

Activity 4

A delivery service charges ₱120 as a fixed fee and ₱18 per kilogram of package weight. Find the cost of delivering a 10-kg package.

Activity 5

A car rental company charges ₱1 500 plus ₱10 per kilometer driven. Find the total cost if the car is driven for 120 kilometers.

Thank you!